Chapter 26: Problems with Solutions

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1. (a) When a battery is connected to the plates of a 3.00-μF capacitor, it stores a charge of 27.0 μC. What is the voltage of the battery? (b) If the same capacitor is connected to another battery and 36.0 μC of charge is stored on the capacitor, what is the voltage of the battery?

(a) From Equation 26.1 for the definition of capacitance, $C = \frac{Q}{\Delta V}$, we have

$$\Delta V = \frac{Q}{C} = \frac{27.0\ \mu C}{3.00\ \mu F} = \boxed{9.00\ V}$$

(b) Similarly,

$$\Delta V = \frac{Q}{C} = \frac{36.0\ \mu C}{3.00\ \mu F} = \boxed{12.0\ V}$$

4. An air-filled spherical capacitor is constructed with **M** inner- and outer-shell radii of 7.00 cm and 14.0 cm, respectively. (a) Calculate the capacitance of the device. (b) What potential difference between the spheres results in a 4.00-μC charge on the capacitor?

(a) For a spherical capacitor with inner radius *a* and outer radius *b*,

$$C = \frac{ab}{k_e(b-a)} = \frac{(0.070\,0\text{ m})(0.140\text{ m})}{(8.99\times10^{9}\text{ N}\cdot\text{m}^2/\text{C}^2)(0.140\text{ m}-0.070\,0\text{ m})}$$

$$= \boxed{15.6\text{ pF}}$$

(b) $$\Delta V = \frac{Q}{C} = \frac{4.00\times10^{-6}\text{ C}}{1.56\times10^{-11}\text{ F}} = 2.57\times10^{5}\text{ V} = \boxed{257\text{ kV}}$$

5. M A 50.0-m length of coaxial cable has an inner conductor that has a diameter of 2.58 mm and carries a charge of 8.10 μC. The surrounding conductor has an inner diameter of 7.27 mm and a charge of $-8.10\ \mu$C. Assume the region between the conductors is air. (a) What is the capacitance of this cable? (b) What is the potential difference between the two conductors?

(a) The capacitance of a cylindrical capacitor is

$$C = \frac{\ell}{2k_e \ln(b/a)}$$

$$= \frac{50.0\ \text{m}}{2\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\ln(7.27\ \text{mm}/2.58\ \text{mm})}$$

$$= \boxed{2.68\ \text{nF}}$$

(b) Method 1: $\Delta V = 2k_e \lambda \ln\left(\frac{b}{a}\right)$

$$\lambda = \frac{Q}{\ell} = \frac{8.10 \times 10^{-6}\ \text{C}}{50.0\ \text{m}} = 1.62 \times 10^{-7}\ \text{C/m}$$

$$\Delta V = 2\left(8.99 \times 10^{9}\ \text{N} \cdot \text{m}^2/\text{C}^2\right)\left(1.62 \times 10^{-7}\ \text{C/m}\right)\ln\left(\frac{7.27\ \text{mm}}{2.58\ \text{mm}}\right)$$

$$= \boxed{3.02\ \text{kV}}$$

Method 2: $\Delta V = \frac{Q}{C} = \frac{8.10 \times 10^{-6}\ \text{C}}{2.68 \times 10^{-9}\ \text{F}} = \boxed{3.02\ \text{kV}}$

7. When a potential difference of 150 V is applied to the plates of a parallel-plate capacitor, the plates carry a surface charge density of 30.0 nC/cm^2. What is the spacing between the plates?

We have $Q = C\Delta V$ and $C = \epsilon_0 A / d$. Thus, $Q = \epsilon_0 A \Delta V / d$

The surface charge density on each plate has the same magnitude, given by

$$\sigma = \frac{Q}{A} = \frac{\epsilon_0 \Delta V}{d}$$

Thus,

$$d = \frac{\epsilon_0 \Delta V}{Q/A} = \frac{\left(8.85 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2}\right)(150\ \mathrm{V})}{\left(30.0 \times 10^{-9}\ \mathrm{C/cm^2}\right)}$$

$$d = \left(4.43 \times 10^{-2}\ \frac{\mathrm{V \cdot C \cdot cm^2}}{\mathrm{N \cdot m^2}}\right)\frac{\left(1\ \mathrm{m^2}\right)}{\left(10^4\ \mathrm{cm^2}\right)}\frac{\mathrm{J}}{\mathrm{V \cdot C}}\frac{\mathrm{N \cdot m}}{\mathrm{J}} = \boxed{4.43\ \mu\mathrm{m}}$$

9. M An air-filled capacitor consists of two parallel plates, each with an area of 7.60 cm^2, separated by a distance of 1.80 mm. A 20.0-V potential difference is applied to these plates. Calculate (a) the electric field between the plates, (b) the surface charge density, (c) the capacitance, and (d) the charge on each plate.

(a) The potential difference between two points in a uniform electric field is $\Delta V = Ed$, so

$$E = \frac{\Delta V}{d} = \frac{20.0\ \text{V}}{1.80\times10^{-3}\ \text{m}} = \boxed{1.11\times10^{4}\ \text{V/m}}$$

(b) The electric field between capacitor plates is $E = \frac{\sigma}{\epsilon_0}$, so $\sigma = \epsilon_0 E$:

$$\sigma = \left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)\left(1.11\times10^{4}\ \text{V/m}\right) = 9.83\times10^{-8}\ \text{C/m}^2$$

$$= \boxed{98.3\ \text{nC/m}^2}$$

(c) For a parallel-plate capacitor, $C = \frac{\epsilon_0 A}{d}$:

$$C = \frac{\left(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2\right)\left(7.60\times10^{-4}\ \text{m}^2\right)}{1.80\times10^{-3}\ \text{m}}$$

$$= 3.74\times10^{-12}\ \text{F} = \boxed{3.74\ \text{pF}}$$

(d) The charge on each plate is $Q = C\Delta V$:

$$Q = \left(3.74\times10^{-12}\ \text{F}\right)(20.0\ \text{V}) = \boxed{74.7\ \text{pC}}$$

11. An isolated, charged conducting sphere of radius 12.0 cm creates an electric field of 4.90×10^4 N/C at a distance 21.0 cm from its center. (a) What is its surface charge density? (b) What is its capacitance?

(a) The electric field outside a spherical charge distribution of radius R is $E = k_e q / r^2$. Therefore,

$$q = \frac{Er^2}{k_e} = \frac{(4.90 \times 10^4 \text{ N/C})(0.210 \text{ m})^2}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2} = 0.240 \ \mu\text{C}$$

Then

$$\sigma = \frac{q}{A} = \frac{0.240 \times 10^{-6} \text{ C}}{4\pi (0.120 \text{ m})^2} = \boxed{1.33 \ \mu\text{C/m}^2}$$

(b) For an isolated charged sphere of radius R,

$$C = 4\pi \epsilon_0 r = 4\pi (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.120 \text{ m}) = \boxed{13.3 \text{ pF}}$$

12. **Review.** A small object of mass m carries a charge q and is suspended by a thread between the vertical plates of a parallel-plate capacitor. The plate separation is d. If the thread makes an angle θ with the vertical, what is the potential difference between the plates?

$$\sum F_y = 0: \quad T\cos\theta - mg = 0$$

$$\sum F_x = 0: \quad T\sin\theta - Eq = 0$$

Dividing, $\tan\theta = \dfrac{Eq}{mg}$,

so $\quad E = \dfrac{mg}{q}\tan\theta$

and $\quad \Delta V = Ed = \boxed{\dfrac{mgd\tan\theta}{q}}$.

13. Two capacitors, $C_1 = 5.00\ \mu\text{F}$ and $C_2 = 12.0\ \mu\text{F}$, are **W** connected in parallel, and the resulting combination is connected to a 9.00-V battery. Find (a) the equivalent capacitance of the combination, (b) the potential difference across each capacitor, and (c) the charge stored on each capacitor.

(a) Capacitors in parallel add. Thus, the equivalent capacitor has a value of

$$C_{eq} = C_1 + C_2 = 5.00\ \mu\text{F} + 12.0\ \mu\text{F} = \boxed{17.0\ \mu\text{F}}$$

(b) The potential difference across each branch is the same and equal to the voltage of the battery.

$$\Delta V = \boxed{9.00\ \text{V}}$$

(c) $$Q_5 = C\Delta V = (5.00\ \mu\text{F})(9.00\ \text{V}) = \boxed{45.0\ \mu\text{C}}$$

$$Q_{12} = C\Delta V = (12.0\ \mu\text{F})(9.00\ \text{V}) = \boxed{108\ \mu\text{C}}$$

14. What If? The two capacitors of Problem 13 ($C_1 = 5.00\ \mu F$ and $C_2 = 12.0\ \mu F$) are now connected in series and to a 9.00-V battery. Find (a) the equivalent capacitance of the combination, (b) the potential difference across each capacitor, and (c) the charge on each capacitor.

(a) In series capacitors add as

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{5.00\ \mu F} + \frac{1}{12.0\ \mu F}$$

$$C_{eq} = \boxed{3.53\ \mu F}$$

(c) We must answer part (c) first before we can answer part (b). The charge on the equivalent capacitor is

$$Q_{eq} = C_{eq}\Delta V = (3.53\ \mu F)(9.00\ V) = 31.8\ \mu C$$

Each of the series capacitors has this same charge on it.

So $Q_1 = Q_2 = \boxed{31.8\ \mu C}$.

21. **M** A group of identical capacitors is connected first in series and then in parallel. The combined capacitance in parallel is 100 times larger than for the series connection. How many capacitors are in the group?

Call C the capacitance of one capacitor and n the number of capacitors. The equivalent capacitance for n capacitors in parallel is

$$C_p = C_1 + C_2 + \cdots + C_n = nC$$

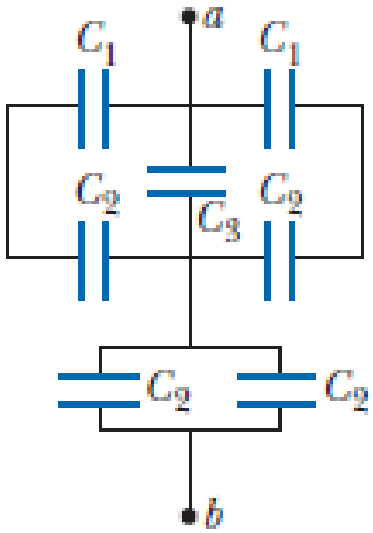
The relationship for n capacitors in series is

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n} = \frac{n}{C}$$

Therefore,

$$\frac{C_p}{C_s} = \frac{nC}{C/n} = n^2 \quad \text{or} \quad n = \sqrt{C_p / C_s} = \sqrt{100} = \boxed{10}$$

22. (a) Find the equivalent capacitance between points a and b for the group of capacitors connected as shown in Figure P26.22. Take $C_1 = 5.00\ \mu\text{F}$, $C_2 = 10.0\ \mu\text{F}$, and $C_3 = 2.00\ \mu\text{F}$. (b) What charge is stored on C_3 if the potential difference between points a and b is 60.0 V?

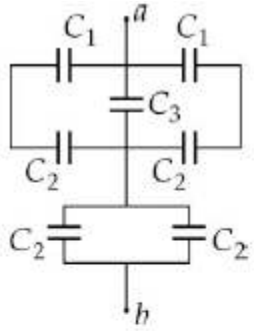


(a) In the upper section, each C_1-C_2 pair, on either side of C_3, are in series:

$$C_s = \left(\frac{1}{5.00} + \frac{1}{10.0}\right)^{-1} = 3.33\ \mu\text{F}$$

and both C_1-C_2 pairs are in parallel to C_3:

$$C_{\text{upper}} = 2(3.33) + 2.00 = 8.67\ \mu\text{F}$$



ANS. FIG. P26.22

In the lower section, the C_2-C_2 pair are in parallel:

$$C_{\text{lower}} = 2(10.0) = 20.0\ \mu\text{F}$$

The upper section is in series with the lower section:

$$C_{\text{eq}} = \left(\frac{1}{8.67} + \frac{1}{20.0}\right)^{-1} = \boxed{6.05\ \mu\text{F}}$$

(b) Capacitors in series carry the same charge as their equivalent capacitor; therefore, the upper section, equivalent to a 8.67-μF capacitor, and the lower section, equivalent to a 20.0-μF capacitor, carry the same charge as a 6.05-μF capacitor:

$$Q_{upper} = Q_{eq} = C_{eq}\Delta V = (6.05\ \mu F)(60.0\ V) \simeq 363\ \mu C$$

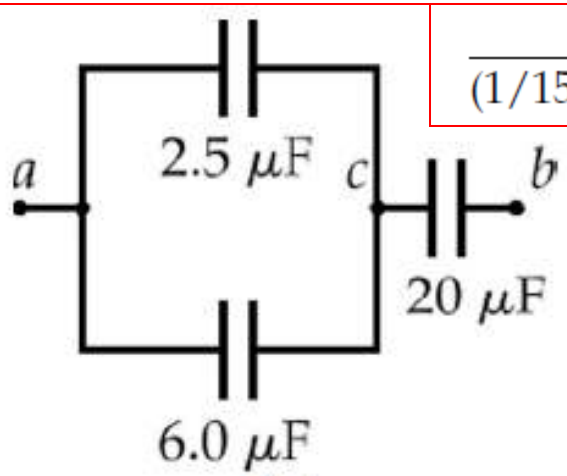
The upper section is equivalent to capacitor C_3 and two 3.33-μF capacitors in parallel, and the voltage across each is the same as that across a 8.67-μF capacitor:

$$\Delta V_{upper} = \frac{Q_{eq}}{C_{eq}} = \frac{363\ \mu C}{8.67\ \mu F} \simeq 41.9\ V$$

Therefore, the charge on C_3 is

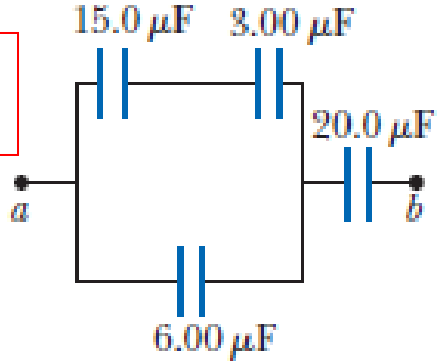
$$Q_3 = C_3\Delta V_3 \simeq (2.00\ \mu F)(41.9\ V) = \boxed{83.7\ \mu C}$$

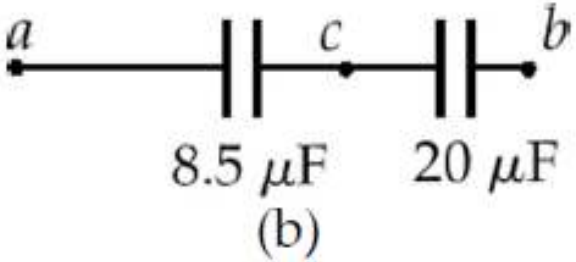
23. **M** Four capacitors are connected as shown in Figure P26.23. (a) Find the equivalent capacitance between points *a* and *b*. (b) Calculate the charge on each capacitor, taking $\Delta V_{ab} = 15.0$ V.



(a)

$$\frac{1}{(1/15.0\ \mu\text{F}) + (1/3.00\ \mu\text{F})} = 2.50\ \mu\text{F}$$





(b)



(c)

$$\frac{1}{(1/8.50\ \mu\text{F}) + (1/20.00\ \mu\text{F})} = \boxed{5.96\ \mu\text{F}}$$

(b) We find the charge on each capacitor and the voltage across each by working backwards through solution figures (c)–(a), alternately applying $Q = C\Delta V$ and $\Delta V = Q/C$ to every capacitor, real or equivalent. For the 5.96-μF capacitor, we have

$$Q = C\Delta V = (5.96\ \mu\text{F})(15.0\ \text{V})$$

$$= \boxed{89.5\ \mu\text{C}}$$

Thus, if *a* is higher in potential than *b*, just 89.5 μC flows between the wires and the plates to charge the capacitors in each picture. In (b) we have, for the 8.5-μF capacitor,

$$\Delta V_{ac} = \frac{Q}{C} = \frac{89.5\ \mu C}{8.50\ \mu F} = 10.5\ V$$

and for the 20.0-μF capacitor in (b), (a), and the original circuit, we have $Q_{20} = 89.5\ \mu C$. Then

$$\Delta V_{cb} = \frac{Q}{C} = \frac{89.5\ \mu C}{20.0\ \mu F} = 4.47\ V$$

Next, the circuit in diagram (a) is equivalent to that in (b), so $\Delta V_{cb} = 4.47$ V and $\Delta V_{ac} = 10.5$ V.

For the 2.50-μF capacitor, $\Delta V = 10.5$ V and

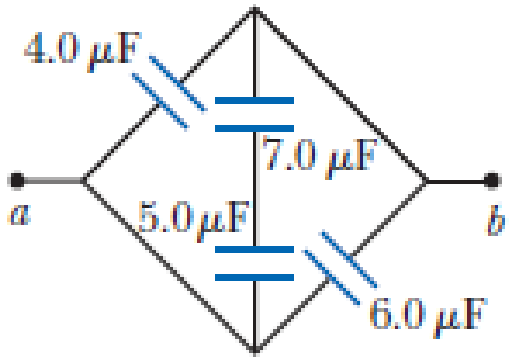
$$Q = C\Delta V = (2.50\ \mu F)(10.5\ V) = \boxed{26.3\ \mu C}$$

For the 6.00-μF capacitor, $\Delta V = 10.5$ V and

$$Q_6 = C\Delta V = (6.00\ \mu F)(10.5\ V) = \boxed{63.2\ \mu C}$$

Now, 26.3 μC having flowed in the upper parallel branch in (a), back in the original circuit we have $Q_{15} = 26.3\ \mu C$ and $Q_3 = 26.3\ \mu C$.

25. Find the equivalent capacitance between points a and b in the combination of capacitors shown in Figure P26.25.



$$C_s = \left(\frac{1}{5.00} + \frac{1}{7.00}\right)^{-1} = 2.92\ \mu\text{F}$$

$$C_p = 2.92 + 4.00 + 6.00 = \boxed{12.9\ \mu\text{F}}$$

27. Two capacitors give an equivalent capacitance of 9.00 pF when connected in parallel and an equivalent capacitance of 2.00 pF when connected in series. What is the capacitance of each capacitor?

$C_p = C_1 + C_2$ and $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$. Substitute $C_2 = C_p - C_1$:

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_p - C_1} = \frac{C_p - C_1 + C_1}{C_1(C_p - C_1)}$$

Simplifying,

$$C_1^2 - C_1 C_p + C_p C_s = 0$$

$$C_1 = \frac{C_p \pm \sqrt{C_p^2 - 4C_p C_s}}{2} = \frac{1}{2}C_p \pm \sqrt{\frac{1}{4}C_p^2 - C_p C_s}$$

$$C_1 = \frac{1}{2}C_p + \sqrt{\frac{1}{4}C_p^2 - C_p C_s}$$

$$= \frac{1}{2}(9.00\text{ pF}) + \sqrt{\frac{1}{4}(9.00\text{ pF})^2 - (9.00\text{ pF})(2.00\text{ pF})}$$

$$= \boxed{6.00\text{ pF}}$$

$$C_2 = C_p - C_1 = \frac{1}{2}C_p - \sqrt{\frac{1}{4}C_p^2 - C_p C_s}$$

$$= \frac{1}{2}(9.00\text{ pF}) - 1.50\text{ pF} = \boxed{3.00\text{ pF}}$$

31. A 12.0-V battery is connected to a capacitor, resulting in 54.0 μC of charge stored on the capacitor. How much energy is stored in the capacitor?

The energy stored in the capacitor is given by

$$U_E = \frac{Q^2}{2C} = \frac{1}{2}Q\Delta V = \frac{1}{2}(54.0\times10^{-6}\ \text{C})(12.0\ \text{V}) = \boxed{3.24\times10^{-4}\ \text{J}}$$

32. (a) A 3.00-μF capacitor is connected to a 12.0-V battery. **W** How much energy is stored in the capacitor? (b) Had the capacitor been connected to a 6.00-V battery, how much energy would have been stored?

(a) $$U_E = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}(3.00\ \mu\text{F})(12.0\ \text{V})^2 = \boxed{216\ \mu\text{J}}$$

(b) $$U_E = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}(3.00\ \mu\text{F})(6.00\ \text{V})^2 = \boxed{54.0\ \mu\text{J}}$$

36. Two identical parallel-plate capacitors, each with capacitance C, are charged to potential difference ΔV and then disconnected from the battery. They are then connected to each other in parallel with plates of like sign connected. Finally, the plate separation in one of the capacitors is doubled. (a) Find the total energy of the system of two capacitors *before* the plate separation is doubled. (b) Find the potential difference across each capacitor *after* the plate separation is doubled. (c) Find the total energy of the system *after* the plate separation is doubled. (d) Reconcile the difference in the answers to parts (a) and (c) with the law of conservation of energy.

Before the capacitors are connected, each has voltage ΔV and charge Q.

(a) Connecting plates of like sign places the capacitors in parallel, so the voltage on each capacitor remains the same.

$$U_{E,\,\text{total}} = \frac{1}{2}C(\Delta V)^2 + \frac{1}{2}C(\Delta V)^2 = \boxed{C(\Delta V)^2}$$

(b) Because $C = \frac{\epsilon_0 A}{d}$, the altered capacitor has new capacitance $C' = \frac{\epsilon_0 A}{2d} = \frac{C}{2}$, and this change in capacitance results in a new potential difference $\Delta V'$ across the parallel capacitors. We can solve for the new potential difference because the total charge remains the same:

$$2Q = C(\Delta V) + C(\Delta V) = C(\Delta V') + \frac{C}{2}(\Delta V') \quad \rightarrow \quad \boxed{\Delta V' = \frac{4\Delta V}{3}}$$

(c) Each capacitor has potential difference $\Delta V'$:

$$U'_{E,\text{total}} = \frac{1}{2}C(\Delta V')^2 + \frac{1}{2}C'(\Delta V')^2 = \frac{1}{2}C\left(\frac{4\Delta V}{3}\right)^2 + \frac{1}{2}\left(\frac{C}{2}\right)\left(\frac{4\Delta V}{3}\right)^2$$

$$= \frac{12}{9}C(\Delta V)^2 = \boxed{4C\frac{(\Delta V)^2}{3}}$$

(d) Positive work is done by the agent pulling the plates apart.

35. Two identical parallel-plate capacitors, each with capacitance 10.0 μF, are charged to potential difference 50.0 V and then disconnected from the battery. They are then connected to each other in parallel with plates of like sign connected. Finally, the plate separation in one of the capacitors is doubled. (a) Find the total energy of the system of two capacitors *before* the plate separation is doubled. (b) Find the potential difference across each capacitor *after* the plate separation is doubled. (c) Find the total energy of the system *after* the plate separation is doubled. (d) Reconcile the difference in the answers to parts (a) and (c) with the law of conservation of energy.

(a) Because the capacitors are connected in parallel, their voltage remains the same:

$$U_E = \frac{1}{2}C(\Delta V)^2 + \frac{1}{2}C(\Delta V)^2 = C(\Delta V)^2$$

$$= \left(10.0\times10^{-6}\ \mu\text{F}\right)(50.0\ \text{V})^2$$

$$= \boxed{2.50\times10^{-2}\ \text{J}}$$

(b) Because $C = \frac{\kappa \in_0 A}{d}$ and $d \rightarrow 2d$, the altered capacitor has new capacitance to $C' = \frac{C}{2}$. The total charge is the same as before:

$$Q_{initial} = Q_{final}$$

$$C(\Delta V) + C(\Delta V) = C(\Delta V') + \frac{C}{2}(\Delta V')$$

$$2C(\Delta V) = \frac{3}{2}C(\Delta V') \quad \rightarrow \quad \Delta V' = \frac{4}{3}\Delta V = \frac{4}{3}(50.0\ \text{V}) = \boxed{66.7\ \text{V}}$$

(c) $$\text{New } U_E' = \frac{1}{2}C(\Delta V')^2 + \frac{1}{2}\left(\frac{1}{2}C\right)(\Delta V')^2 = \frac{3}{4}C(\Delta V')^2 = \frac{3}{4}C\left(\frac{4\Delta V}{3}\right)^2$$

$$U_E' = \frac{4}{3}C(\Delta V)^2 = \frac{4}{3}U_E = \frac{4}{3}\left(2.50\times10^{-2}\ \text{J}\right) = \boxed{3.30\times10^{-2}\ \text{J}}$$

(d) Positive work is done by the agent pulling the plates apart.

45. Determine (a) the capacitance and (b) the maximum potential difference that can be applied to a Teflon-filled parallel-plate capacitor having a plate area of $1.75\ \text{cm}^2$ and a plate separation of 0.040 0 mm.

(a) $$C = \frac{\kappa \epsilon_0 A}{d} = \frac{2.10\left(8.85\times10^{-12}\ \text{F/m}\right)\left(1.75\times10^{-4}\ \text{m}^2\right)}{4.00\times10^{-5}\ \text{m}} = 8.13\times10^{-11}\ \text{F}$$

$$= \boxed{81.3\ \text{pF}}$$

(b) $$\Delta V_{max} = E_{max} d = \left(60.0\times10^{6}\ \text{V/m}\right)\left(4.00\times10^{-5}\ \text{m}\right) = \boxed{2.40\ \text{kV}}$$

50. M A small, rigid object carries positive and negative 3.50-nC charges. It is oriented so that the positive charge has coordinates (−1.20 mm, 1.10 mm) and the negative charge is at the point (1.40 mm, −1.30 mm). (a) Find the electric dipole moment of the object. The object is placed in an electric field $\vec{\mathbf{E}} = (7.80 \times 10^3\,\hat{\mathbf{i}} - 4.90 \times 10^3\,\hat{\mathbf{j}})$ N/C. (b) Find the torque acting on the object. (c) Find the potential energy of the object–field system when the object is in this orientation. (d) Assuming the orientation of the object can change, find the difference between the maximum and minimum potential energies of the system.

(a) The displacement from negative to positive charge is

$$2\vec{a} = \left(-1.20\hat{i} + 1.10\hat{j}\right)\ \text{mm} - \left(1.40\hat{i} - 1.30\hat{j}\right)\ \text{mm}$$

$$= \left(-2.60\hat{i} + 2.40\hat{j}\right) \times 10^{-3}\ \text{m}$$

The electric dipole moment is $\vec{p} = 2\vec{a}q$

$$\vec{p} = \left(3.50 \times 10^{-9}\ \text{C}\right)\left(-2.60\hat{i} + 2.40\hat{j}\right) \times 10^{-3}\ \text{m}$$

$$= \boxed{\left(-9.10\hat{i} + 8.40\hat{j}\right) \times 10^{-12}\ \text{C} \cdot \text{m}}$$

(b) The torque exerted by the field on the dipole is

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$= \left[\left(-9.10\hat{i} + 8.40\hat{j}\right) \times 10^{-12}\ \text{C} \cdot \text{m}\right] \times \left[\left(7.80\hat{i} - 4.90\hat{j}\right) \times 10^{3}\ \text{N/C}\right]$$

$$= \left(+44.6\hat{k} - 65.5\hat{k}\right) \times 10^{-9}\ \text{N} \cdot \text{m} = \boxed{-2.09 \times 10^{-8}\hat{k}\ \text{N} \cdot \text{m}}$$

(c) Relative to zero energy when it is perpendicular to the field, the dipole has potential energy

$$U = -\vec{p} \cdot \vec{E}$$

$$= -\left[\left(-9.10\hat{i} + 8.40\hat{j}\right) \times 10^{-12}\ \text{C} \cdot \text{m}\right] \cdot \left[\left(7.80\hat{i} - 4.90\hat{j}\right) \times 10^{3}\ \text{N/C}\right]$$

$$= (71.0 + 41.2) \times 10^{-9}\ \text{J} = \boxed{112\ \text{nJ}}$$

(d) For convenience we compute the magnitudes

$$|\vec{p}| = \sqrt{(9.10)^2 + (8.40)^2} \times 10^{-12}\ \text{C}\cdot\text{m} = 12.4 \times 10^{-12}\ \text{C}\cdot\text{m}$$

and $$|\vec{E}| = \sqrt{(7.80)^2 + (4.90)^2} \times 10^3\ \text{N/C} = 9.21 \times 10^3\ \text{N/C}$$

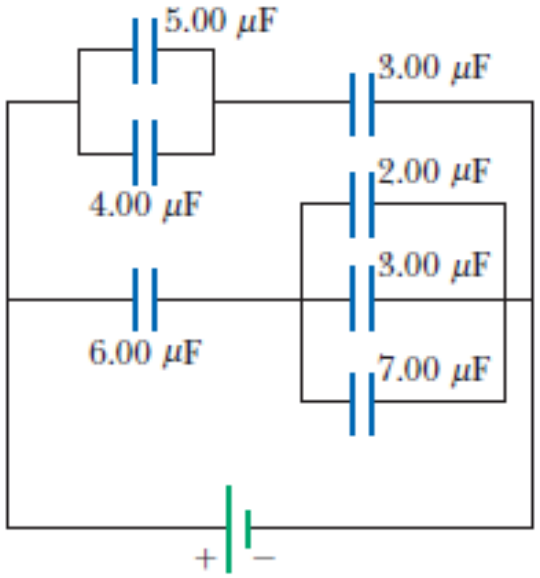
The maximum potential energy occurs when the dipole moment is opposite in direction to the field, and is

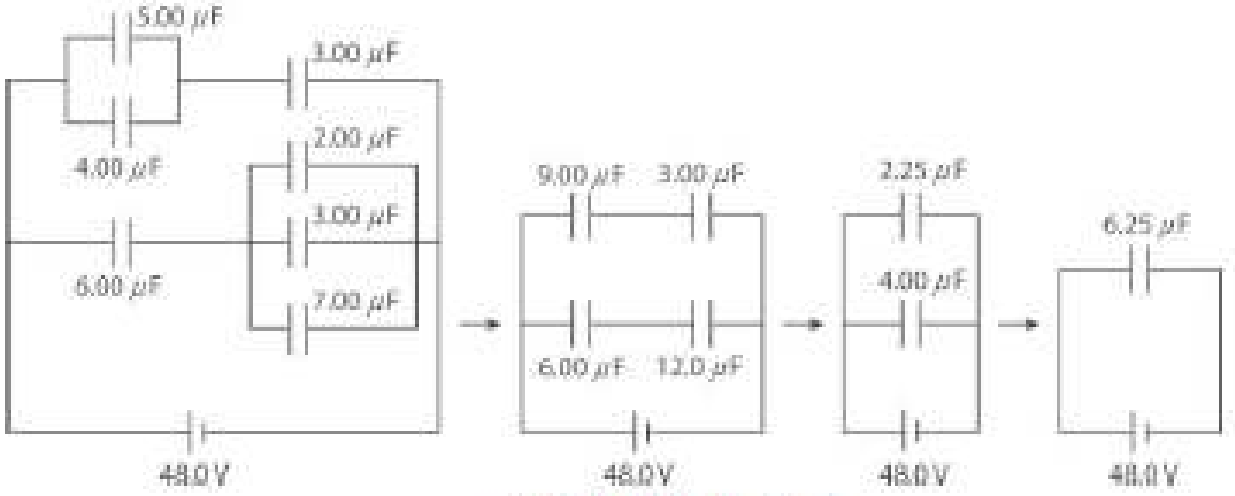
$$U_{max} = -\vec{p}\cdot\vec{E} = -|\vec{p}||\vec{E}|(-1) = |\vec{p}||\vec{E}| = 114\ \text{nJ}$$

The minimum potential energy configuration is the stable equilibrium position with the dipole aligned with the field. The value is $U_{min} = -114$ nJ

Then the difference, representing the range of potential energies available to the dipole, is $U_{max} - U_{min} = \boxed{228\ \text{nJ}}$.

54. Find the equivalent capacitance of the group of capacitors shown in Figure P26.54.





ANS. FIG. P26.54

Thus, $C_{eq} = \boxed{6.25\ \mu F}$

63. A 10.0-μF capacitor is charged to 15.0 V. It is next connected in series with an uncharged 5.00-μF capacitor. The series combination is finally connected across a 50.0-V battery as diagrammed in Figure P26.63. Find the new potential differences across the 5.00-μF and 10.0-μF capacitors after the switch is thrown closed.

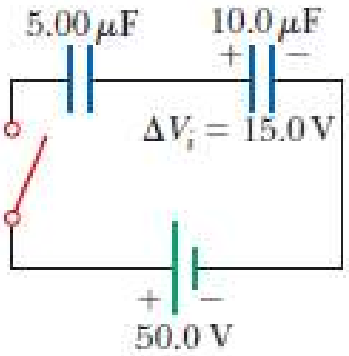


Figure P26.63

The initial charge on the larger capacitor is

$$Q = C\Delta V = (10.0\ \mu F)(15.0\ V) = 150\ \mu C$$

An additional charge q is pushed through the 50.0-V battery, giving the smaller capacitor charge q and the larger charge $150\ \mu C + q$.

Then
$$50.0\ V = \frac{q}{5.00\ \mu F} + \frac{150\ \mu C + q}{10.0\ \mu F},$$

$$500\ \mu C = 2q + 150\ \mu C + q$$

$$q = 117\ \mu C$$

So across the 5.00-μF capacitor,

$$\Delta V = \frac{q}{C} = \frac{117\ \mu C}{5.00\ \mu F} = \boxed{23.3\ V}$$

Across the 10.0-μF capacitor,

$$\Delta V = \frac{150\ \mu C + 117\ \mu C}{10.0\ \mu F} = \boxed{26.7\ V}$$

75. Determine the equivalent capacitance of the combination shown in Figure P26.75. *Suggestion:* Consider the symmetry involved.

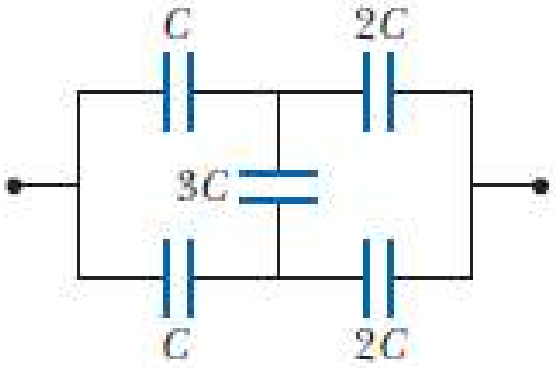
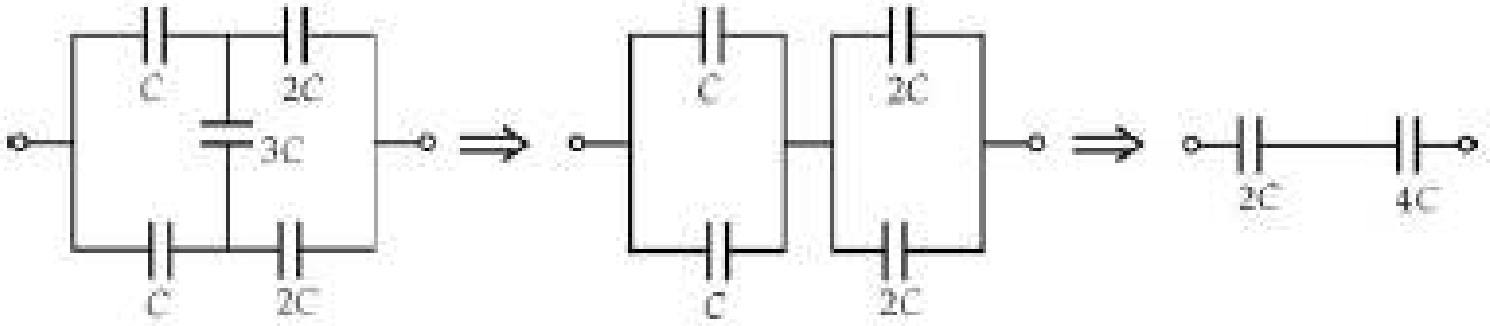


Figure P26.75

P26.75 By symmetry, the potential difference across $3C$ is zero, so the circuit reduces to (see ANS. FIG. P26.75):

$$C_{eq} = \left(\frac{1}{2C} + \frac{1}{4C}\right)^{-1} = \frac{8}{6}C = \boxed{\frac{4}{3}C}$$



77. Calculate the equivalent capacitance between points a and b in Figure P26.77. Notice that this system is not a simple series or parallel combination. *Suggestion:* Assume a potential difference ΔV between points a and b. Write expressions for ΔV_{ab} in terms of the charges and capacitances for the various possible pathways from a to b and require conservation of charge for those capacitor plates that are connected to each other.

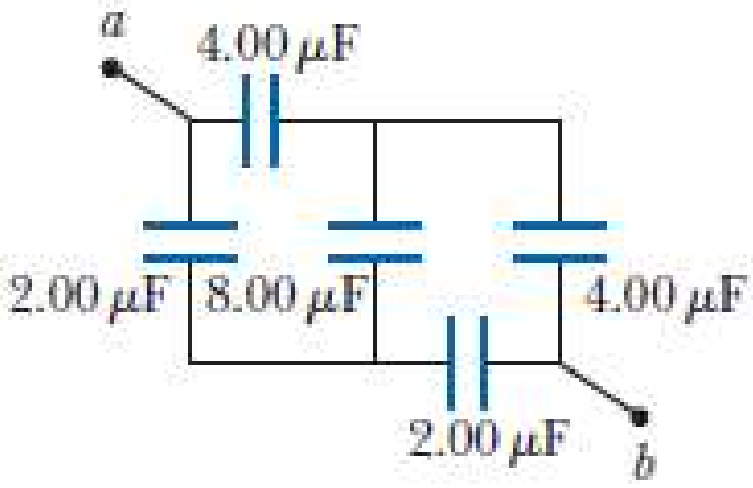


Figure P26.77

Assume a potential difference across *a* and *b*, and notice that the potential difference across 8.00 μF the capacitor must be zero by symmetry. Then the equivalent capacitance can be determined from the circuit shown in ANS. FIG. P26.77, and is $C_{ab} = \boxed{3.00\ \mu F}$.



ANS. FIG. P26.77

Example 26.10 A Partially Filled Capacitor

A parallel-plate capacitor with a plate separation d has a capacitance C_0 in the absence of a dielectric. What is the capacitance when a slab of dielectric material of dielectric constant κ and thickness $\frac{1}{3}d$ is inserted between the plates (Fig. 26.29a)?

Solution In Example 26.9, we found that we could insert a metallic slab between the plates of a capacitor and consider the combination as two capacitors in series. The resulting capacitance was independent of the location of the slab. Furthermore, if the thickness of the slab approaches zero,

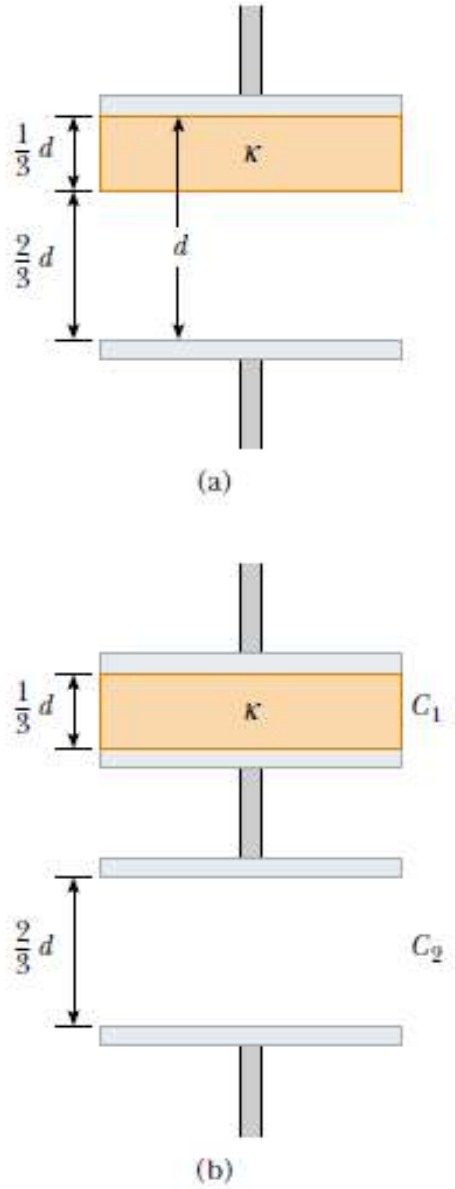


Figure 26.29 (Example 26.10) (a) A parallel-plate capacitor of plate separation d partially filled with a dielectric of thickness $d/3$. (b) The equivalent circuit of the capacitor consists of two capacitors connected in series.

then the capacitance of the system approaches the capacitance when the slab is absent. From this, we conclude that we can insert an infinitesimally thin metallic slab anywhere between the plates of a capacitor without affecting the capacitance. Thus, let us imagine sliding an infinitesimally thin metallic slab along the bottom face of the dielectric shown in Figure 26.29a. We can then consider this system to be the series combination of the two capacitors shown in Figure 26.29b: one having a plate separation $d/3$ and filled with a dielectric, and the other having a plate separation $2d/3$ and air between its plates.

From Equations 26.15 and 26.3, the two capacitances are

$$C_1 = \frac{\kappa \epsilon_0 A}{d/3} \qquad \text{and} \qquad C_2 = \frac{\epsilon_0 A}{2d/3}$$

Using Equation 26.10 for two capacitors combined in series, we have

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d/3}{\kappa \epsilon_0 A} + \frac{2d/3}{\epsilon_0 A}$$

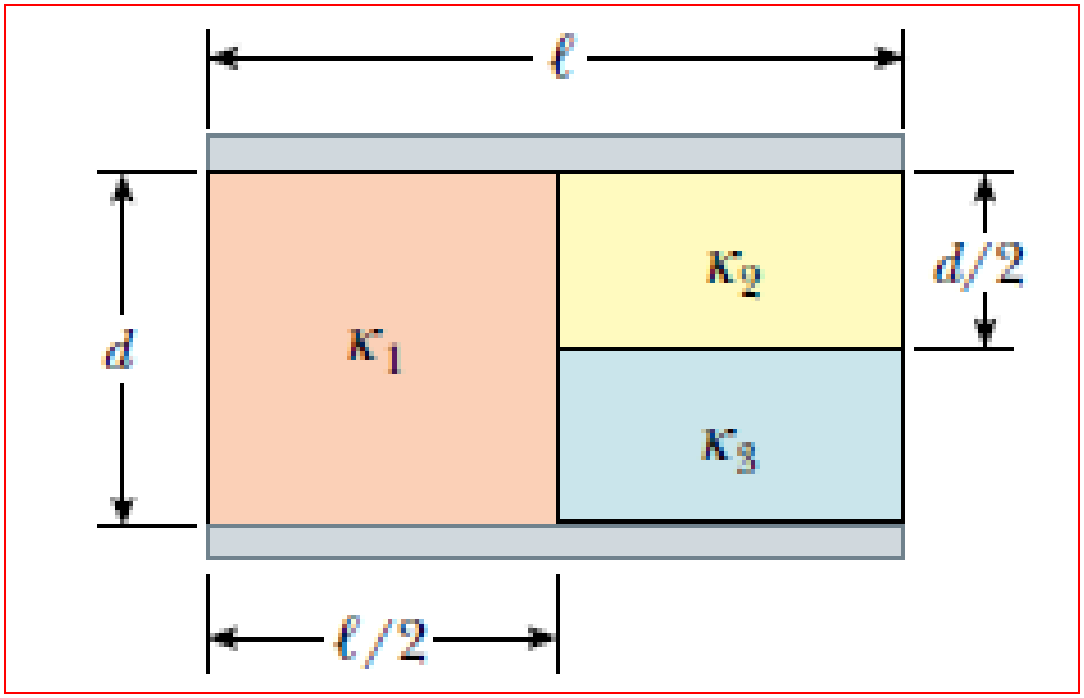
$$\frac{1}{C} = \frac{d}{3\epsilon_0 A}\left(\frac{1}{\kappa} + 2\right) = \frac{d}{3\epsilon_0 A}\left(\frac{1 + 2\kappa}{\kappa}\right)$$

$$C = \left(\frac{3\kappa}{2\kappa + 1}\right)\frac{\epsilon_0 A}{d}$$

Because the capacitance without the dielectric is $C_0 = \epsilon_0 A/d$, we see that

$$C = \left(\frac{3\kappa}{2\kappa + 1}\right) C_0$$

A parallel-plate capacitor is constructed by filling the space between two square plates with blocks of three dielectric materials, as in Figure P26.61. You may assume that $\ell \gg d$. (a) Find an expression for the capacitance of the device in terms of the plate area A and d, κ_1, κ_2, and κ_3. (b) Calculate the capacitance using the values $A = 1.00\ \text{cm}^2$, $d = 2.00\ \text{mm}$, $\kappa_1 = 4.90$, $\kappa_2 = 5.60$, and $\kappa_3 = 2.10$.



(a) $$C_1 = \frac{\kappa_1 \epsilon_0 A/2}{d}\,;\; C_2 = \frac{\kappa_2 \epsilon_0 A/2}{d/2}\,;\; C_3 = \frac{\kappa_3 \epsilon_0 A/2}{d/2}$$

$$\left(\frac{1}{C_2}+\frac{1}{C_3}\right)^{-1} = \frac{C_2 C_3}{C_2+C_3} = \frac{\epsilon_0 A}{d}\left(\frac{\kappa_2\kappa_3}{\kappa_2+\kappa_3}\right)$$

$$C = C_1 + \left(\frac{1}{C_2}+\frac{1}{C_3}\right)^{-1} = \boxed{\frac{\epsilon_0 A}{d}\left(\frac{\kappa_1}{2}+\frac{\kappa_2\kappa_3}{\kappa_2+\kappa_3}\right)}$$

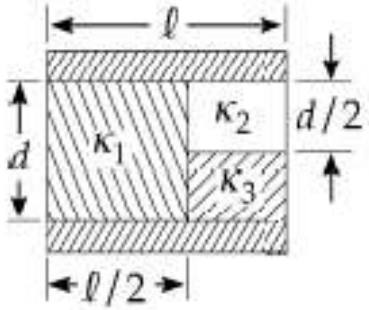


FIG. P26.61

(b) Using the given values we find: $C_{\text{total}} = 1.76 \times 10^{-12}\ \text{F} = \boxed{1.76\ \text{pF}}$.